

Probability

Definition 1.0.1 ► Discrete Probability Model

Let Ω be a set, $|\Omega| < \infty$. $\mathcal{Q} = \varphi(\Omega)$ the collection of all subsets of Ω . $P : \varphi(\Omega) \rightarrow [0, 1]$ a function. If

1. $P(\Omega) = 1$.
 2. $P(A \cup B) = P(A) + P(B)$ for all $A, B \in \varphi(\Omega)$, $A \cap B = \emptyset$.
- Then we call $(\Omega, \mathcal{Q}, P)$ a discrete probability model.

Proposition 1.0.2

Let $(\Omega, \mathcal{Q}, P)$ be a discrete probability model,

1. Let $A \in \varphi(\Omega)$, $\bar{A} := \{x \in \Omega, x \notin A\}$. Then

$$P(\bar{A}) = 1 - P(A)$$

2. $A, B \in \varphi(\Omega)$, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Remark.

If $P(\{\cdot\}) = \frac{1}{|\Omega|}$, then we say the probability is **uniform**.

Example 1.0.3

Take $\Omega = \{a, b\}$, $\mathcal{Q} = \varphi(\Omega) = \{\emptyset, \{a\}, \{b\}, \Omega\}$. $P(\{a\}) = \theta, P(\{b\}) = 1 - \theta$. Then P can extend to a probability, and $(\Omega, \mathcal{Q}, P)$ comes to a discrete probability model.

Definition 1.0.4 ► Random Variable

A **Random Variable** (R.V) is just a map $X : \Omega \rightarrow \mathbb{R}$.

Example 1.0.5 ▶ A Random Variable

$A \in \mathcal{F}(\Omega)$. Let $X = \chi_A$. Then X is a random variable. And

$$P(X = 1) = P(A), \quad P(X = 0) = 1 - P(A).$$

Definition 1.0.6 ▶ Expectation (Mean Value of a R.V)

Let X be a R.V. If

1. $|X(\Omega)| < \infty$,